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# Theoretical Performance Analysis of the General Transfer Function GSC

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## ABSTRACT

In this paper we present a theoretical analysis of the newly suggested TF-GSC algorithm. In recent work we considered a microphone array located in a reverberated room, where general transfer functions (TFs) relate the source signal and the microphones. The array is used for enhancing a speech signal contaminated by interference. The new, frequency-domain algorithm enables dealing with a complicated TF in the same simple manner as Griffiths and Jim GSC algorithm deals with delay-only arrays. It was shown that it is sufficient to use the ratio between the different TFs rather than the TFs themselves in order to implement the suggested algorithm. An unbiased estimate of the TFs ratios was achieved by exploiting the non-stationarity of the speech signal. Experimental results show that the algorithm is working well in various real-life scenarios and is outperforming the classical GSC algorithm.

In this contribution we turn to analytical evaluation of the performance of the suggested algorithm. We first derive a general expression of the enhancer output. We use this expression to evaluate two figures of merit, i.e. noise reduction ability and the amount of imposed distortion. The expression depends on the TFs involved, the noise field and the quality of estimation of the TFs ratios. For the latter dependency, we provide an approximated analysis of the signal's non-stationarity based estimation procedure and explore its dependency on the actual speech signal and the SNR level. It is shown, that for SNR levels as low as -5dB the estimation is almost bias free and imposes only minor distortion. This analysis is with good correspondence with previously achieved experimental results.